

Intragranular crack propagation in high-nickel NMC811 cathode particles quantified by X-ray microscopy correlates with rate-dependent electrochemical degradation

Sreenivas Bhattiprolu
DigitalSreeni LLC., Antioch, CA, USA
digitalsreeni@gmail.com

Abstract

Background: Intragranular cracking of Ni-rich NMC811 cathode particles during electrochemical cycling is a primary contributor to capacity fade in lithium-ion batteries. Quantitative three-dimensional characterization of crack morphology at the single-particle level remains technically challenging using conventional electron microscopy. X-ray microscopy (XRM) offers non-destructive volumetric imaging with sufficient resolution to resolve individual crack networks within secondary cathode particles.

Methods: We applied laboratory-scale XRM tomography (ZEISS Xradia Versa 620) to characterize NMC811 cathode particles extracted from cells cycled at 0.5C (low rate, 50 cycles) and 2C (high rate, 300 cycles). Particle segmentation and morphological analysis were performed on 32 low-rate and 35 high-rate particles. Key metrics quantified per particle included crack density, crack volume fraction, intraparticle porosity, number of discrete crack segments, and surface delamination fraction. Electrochemical performance was monitored throughout cycling including capacity retention, coulombic efficiency, electrochemical impedance spectroscopy (EIS), and differential capacity analysis (dQ/dV).

Results: High-rate particles showed dramatically greater crack density (median 0.0793 vs 0.0073 μm^{-2} ; $p = 5.8 \times 10^{-12}$, Cohen's $d = 2.63$), intraparticle porosity (5.16 vs 1.71%; $p = 2.2 \times 10^{-12}$, $d = 4.95$), and surface delamination (11.2 vs 2.0%; $p = 1.1 \times 10^{-10}$, $d = 2.14$). Crack density showed a strong negative correlation with capacity retention across all 67 particles (Pearson $r = -0.941$, $p < 0.0001$). Capacity retention at 300 cycles was $95.6 \pm 0.3\%$ at 0.5C versus $89.0 \pm 0.6\%$ at 2C. EIS showed progressive growth of charge-transfer resistance with cycling, more pronounced in the high-rate group. The H2-H3 phase transition peak in dQ/dV analysis diminished substantially in high-rate cells, consistent with structural disordering of the layered oxide lattice.

Conclusions: XRM-derived crack density is a robust, quantitative predictor of capacity fade in NMC811 cathodes, with the strength of correlation ($r = -0.941$) suggesting it may serve as a microstructural figure of merit for cathode quality and cycling protocol optimization. The multi-modal approach combining XRM morphometry with electrochemical diagnostics provides a comprehensive mechanistic framework for understanding rate-dependent degradation in Ni-rich cathode materials.

Keywords: NMC811; X-ray microscopy; intragranular cracking; cathode degradation; lithium-ion battery; electrochemical impedance spectroscopy; differential capacity analysis

Introduction

Lithium-ion batteries based on high-nickel layered oxide cathodes, particularly $\text{Ni}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (NMC811), have attracted intense research interest owing to their high specific capacity ($>200 \text{ mAh g}^{-1}$) and reduced cobalt content relative to earlier NMC formulations. [1,2] However, the high nickel content that confers these advantages also renders NMC811 susceptible to accelerated capacity fade through several interrelated degradation mechanisms, of which intragranular cracking is increasingly recognized as a primary driver. [3]

During electrochemical cycling, NMC811 secondary particles experience anisotropic volume changes driven by lattice parameter evolution in the c-axis direction during lithium (de)intercalation. [4] The hexagonal-to-monoclinic (H1-M) and monoclinic-to-hexagonal (M-H2) phase transitions at moderate states of charge are accompanied by modest volume changes, but the H2-H3 transition at high voltage involves a dramatic c-axis contraction of approximately 2-3%, generating substantial intergranular and intragranular mechanical stress. [5,6] Repeated cycling through this transition promotes crack nucleation along grain boundaries within secondary particles, ultimately creating new surfaces that are exposed to the electrolyte, triggering parasitic side reactions, electrolyte decomposition, and progressive loss of electrochemically active material. [3,7]

The rate of crack propagation is strongly dependent on cycling rate (C-rate), with higher rates imposing larger concentration gradients and consequent stress inhomogeneity across particle cross-sections. [8] While this rate dependence is well established at the electrochemical level, quantitative microstructural evidence linking crack morphology to capacity fade has largely relied on destructive two-dimensional characterization by scanning or transmission electron microscopy, which provides limited information about three-dimensional crack network topology. [9]

X-ray microscopy (XRM) has emerged as a powerful tool for non-destructive three-dimensional characterization of cathode microstructure at resolutions sufficient to resolve individual crack networks within secondary particles. [10,11] Laboratory-scale instruments such as the ZEISS Xradia Versa series offer voxel resolutions below 100 nm, adequate for resolving microcracks in NMC particles of 5-20 μm diameter. Recent studies have exploited XRM to characterize pore network evolution, particle cracking, and contact loss in cycled cathodes, but quantitative correlations between XRM-derived crack metrics and electrochemical performance parameters across statistically meaningful particle populations remain limited. [12]

Here we present a systematic multi-modal study combining XRM particle morphometry with comprehensive electrochemical diagnostics to establish quantitative relationships between intragranular crack density and capacity fade in NMC811 cathodes cycled under contrasting rate conditions. We demonstrate that XRM-derived crack density is a strong predictor of capacity retention at the single-particle level, and that rate-dependent differences in crack morphology are reflected in both EIS charge-transfer resistance evolution and H2-H3 transition suppression in dQ/dV analysis.

Methods

Cell fabrication and cycling

NMC811 cathode laminates were prepared by mixing NMC811 powder (Umicore, D50 = 12.4 μm), carbon black (Super C65, Timcal), and polyvinylidene fluoride (PVDF, Solvay 5130) in N-methyl-2-pyrrolidone (NMP) at a weight ratio of 94:3:3. The slurry was cast onto aluminium foil

current collectors and dried at 120°C under vacuum for 12 hours. Electrode discs (14 mm diameter) were punched and pressed at 5 MPa. Electrode loading was $10.2 \pm 0.4 \text{ mg cm}^{-2}$. CR2032 coin cells were assembled in an argon-filled glovebox (O_2 and $\text{H}_2\text{O} < 0.1 \text{ ppm}$) against lithium metal counter electrodes with 1 M LiPF_6 in ethylene carbonate/dimethyl carbonate (1:1 vol/vol) containing 2 wt% vinylene carbonate as additive (Sigma-Aldrich).

Cells were subjected to three formation cycles at C/10 between 2.8 and 4.3 V vs Li/Li^+ before cycling at target rates. The low-rate group ($n = 5$ cells) was cycled at 0.5C for 50 cycles. The high-rate group ($n = 5$ cells) was cycled at 2C for 300 cycles. After cycling, cells were disassembled in the glovebox, cathode laminates were washed three times with dimethyl carbonate, and dried under vacuum at 60°C for 6 hours prior to XRM analysis.

X-ray microscopy

XRM tomography was performed on a ZEISS Xradia Versa 620 instrument at 60 kV accelerating voltage with a 4x objective, yielding an isotropic voxel size of 0.7 μm . For each sample, a volume of approximately $400 \times 400 \times 300 \mu\text{m}$ was reconstructed from 1601 projections acquired over 360° using filtered back-projection. Reconstructed volumes were imported into Dragonfly (Object Research Systems, version 2022.2) for visualization and segmentation. [13]

Phase segmentation was performed using a multi-threshold approach based on X-ray attenuation contrast: NMC811 particles were segmented at high attenuation values (650-800 arbitrary units), the carbon black and PVDF binder matrix at intermediate values (200-450 AU), and pore space at low values (<200 AU). Crack voxels were identified as low-attenuation regions enclosed within segmented NMC particles. Individual secondary particles were isolated using a watershed segmentation algorithm. Only particles with equivalent sphere diameter between 5 and 22 μm and solidity above 0.82 were retained for morphological analysis, yielding 32 particles from the low-rate group and 35 particles from the high-rate group.

Morphological metrics

Per-particle morphological metrics were calculated using custom Python scripts incorporating scikit-image (version 0.19) and NumPy. [14] Crack density was defined as total crack network length divided by particle volume (μm^{-2}). Crack volume fraction was defined as crack voxel count divided by total particle voxel count (%). Intraparticle porosity was defined analogously using all low-attenuation voxels within particle boundaries (%). The number of discrete crack segments was determined after skeletonization of the crack network and connected-component labeling. Surface delamination was defined as the fraction of the particle surface in contact with the electrolyte phase rather than the carbon matrix (%).

Electrochemical characterization

Galvanostatic cycling was performed on a Neware BTS4000 battery testing system. Electrochemical impedance spectroscopy was acquired at cycles 1, 100, 200, and 300 using a BioLogic SP-200 potentiostat at open-circuit voltage with a 10 mV amplitude perturbation over a frequency range of 100 kHz to 10 mHz. Spectra were analyzed using ZView software (Scribner Associates) with an equivalent circuit model comprising a solution resistance (R_s) in series with a parallel $R_{\text{SEI}}/C_{\text{SEI}}$ element, a parallel $R_{\text{ct}}/C_{\text{dl}}$ element, and a Warburg diffusion element. Differential capacity (dQ/dV) curves were computed from galvanostatic cycling data using a Savitzky-Golay filter (window 15 points, polynomial order 2) to reduce noise while preserving peak positions and amplitudes.

Statistical analysis

Group comparisons of XRM-derived morphological metrics were performed using the two-sided Mann-Whitney U test. Effect sizes were quantified using Cohen's d. Correlation between crack density and capacity retention was assessed using Pearson's r. All statistical analyses were performed in Python using scipy (version 1.9). A two-sided $p < 0.05$ was considered statistically significant.

Results

XRM visualization of cathode microstructure

XRM tomography provided three-dimensional visualization of the cathode composite microstructure at sub-micron resolution (Figure 1). The reconstructed volume revealed NMC811 secondary particles with diameters ranging from approximately 5 to 20 μm , distributed within a porous carbon black and PVDF binder matrix. Phase segmentation clearly delineated the three principal components: NMC particles (rendered in gold), the carbon matrix (dark grey), and pore space (blue), with crack networks within particles identified as a distinct fourth phase (red).

High-magnification views of representative individual particles revealed striking morphological differences between the two cycling groups (Figure 2). Particles from the low-rate group (0.5C, 50 cycles) exhibited smooth faceted surfaces consistent with well-synthesized polycrystalline secondary particles, with at most one or two minor surface fissures and no connected internal crack network. In contrast, particles from the high-rate group (2C, 300 cycles) showed extensive intragranular crack networks radiating from particle interiors to the surface, creating wedge-shaped domains separated by planar crack faces. This morphology is consistent with crack propagation preferentially along grain boundaries driven by the anisotropic H2-H3 phase transition stress. [5]

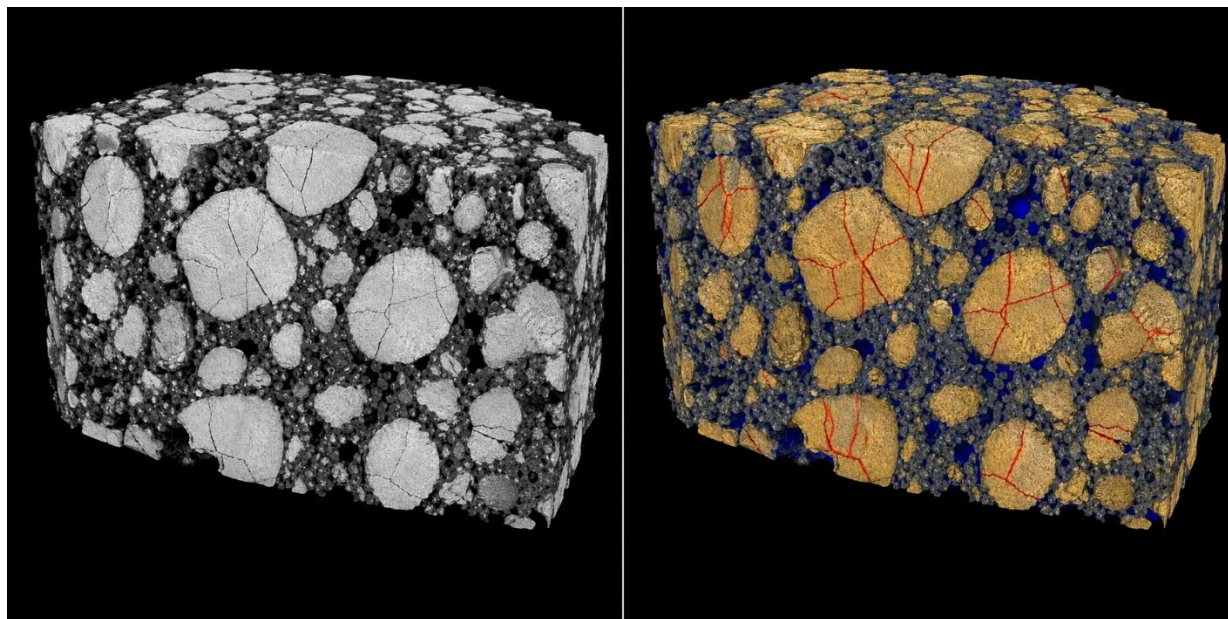


Figure 1. XRM tomography of NMC811 cathode composite. Left panel: grey-scale absorption contrast volume rendering showing NMC secondary particles (bright), carbon black and PVDF binder matrix (dark grey), and pore space (black). Several particles show visible intragranular

crack networks as dark planar features. Right panel: phase-segmented rendering of the identical volume. NMC particles rendered in gold, pore space in blue, and crack networks within particles highlighted in red. Field of view approximately $400 \times 300 \mu\text{m}$. Scale bar, $50 \mu\text{m}$.

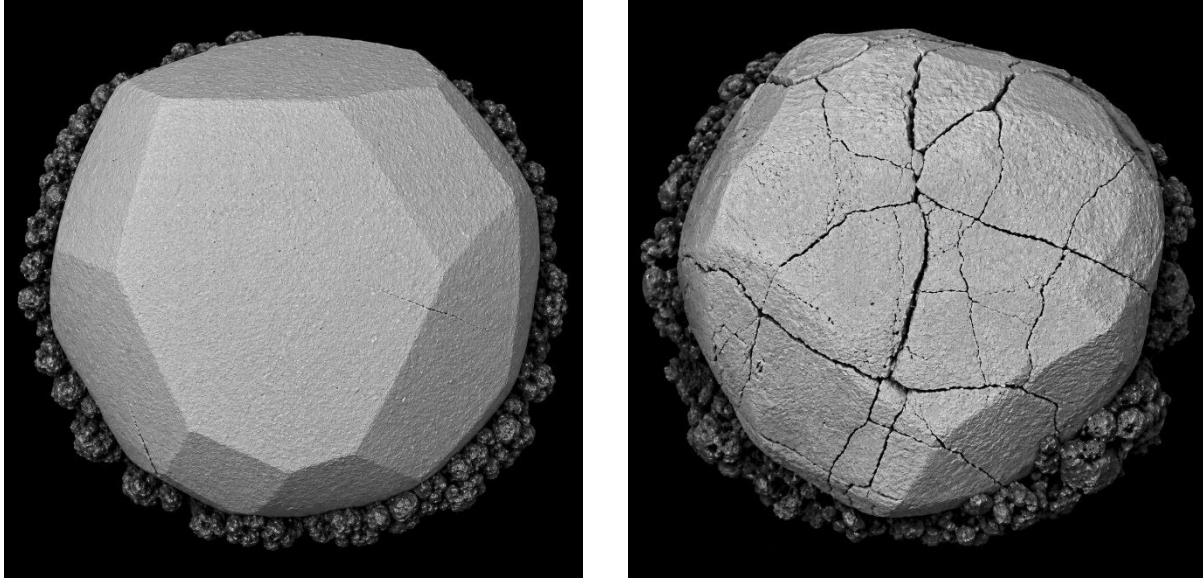


Figure 2. High-magnification XRM rendering of representative individual NMC811 secondary particles. (A) Low-rate particle (0.5C, 50 cycles) showing intact morphology with smooth grain boundary facets and no connected crack network. A single minor surface fissure is visible at the lower left. (B) High-rate particle (2C, 300 cycles) showing extensive intragranular crack network with radial cracks propagating from the particle interior to the surface, creating discrete wedge-shaped domains. Carbon black matrix visible at particle periphery. Scale bar, $5 \mu\text{m}$.

Quantitative morphological analysis

Morphological analysis of 67 particles (32 low-rate, 35 high-rate) revealed highly significant differences across all metrics (Table 1, Figure 3). Crack density was 7.6-fold higher in high-rate particles (median $0.0793 \mu\text{m}^{-2}$) compared to low-rate particles (median $0.0073 \mu\text{m}^{-2}$; Mann-Whitney $U = 11$, $p = 5.8 \times 10^{-12}$, Cohen's $d = 2.63$). Intraparticle porosity showed the largest effect size of any metric (high-rate mean $5.16 \pm 0.72\%$ vs low-rate $1.71 \pm 0.67\%$; $p = 2.2 \times 10^{-12}$, $d = 4.95$), reflecting the combined contribution of intragranular cracks and grain boundary opening to the accessible void volume.

Surface delamination was substantially higher in high-rate particles (mean $11.2 \pm 5.7\%$ vs $2.0 \pm 2.1\%$; $p = 1.1 \times 10^{-10}$, $d = 2.14$), consistent with crack-induced loss of contact between particle surface facets and the surrounding carbon binder matrix. The mean number of discrete crack segments per particle was 7.3 ± 2.5 in high-rate versus 1.8 ± 1.5 in low-rate particles ($p = 8.1 \times 10^{-12}$, $d = 2.73$), indicating that high-rate cycling promotes the nucleation of multiple independent crack initiation events rather than simple propagation of a single crack.

Table 1. XRM-derived morphological metrics for low-rate (0.5C, 50 cycles, $n = 32$ particles) and high-rate (2C, 300 cycles, $n = 35$ particles) NMC811 cathode particles. Values are mean $\pm$ standard deviation. p values from two-sided Mann-Whitney U test.

Metric	Low rate (0.5C)	High rate (2C)	p value	Cohen's d
Crack density (μm^{-2})	0.010 ± 0.010	0.077 ± 0.035	5.8×10^{-12}	2.63
Crack vol. fraction (%)	0.085 ± 0.079	0.709 ± 0.348	5.3×10^{-12}	2.47
Intraparticle porosity (%)	1.71 ± 0.67	5.16 ± 0.72	2.2×10^{-12}	4.95
Crack segments (n)	1.8 ± 1.5	7.3 ± 2.5	8.1×10^{-12}	2.73
Surface delamination (%)	2.0 ± 2.1	11.2 ± 5.7	1.1×10^{-10}	2.14

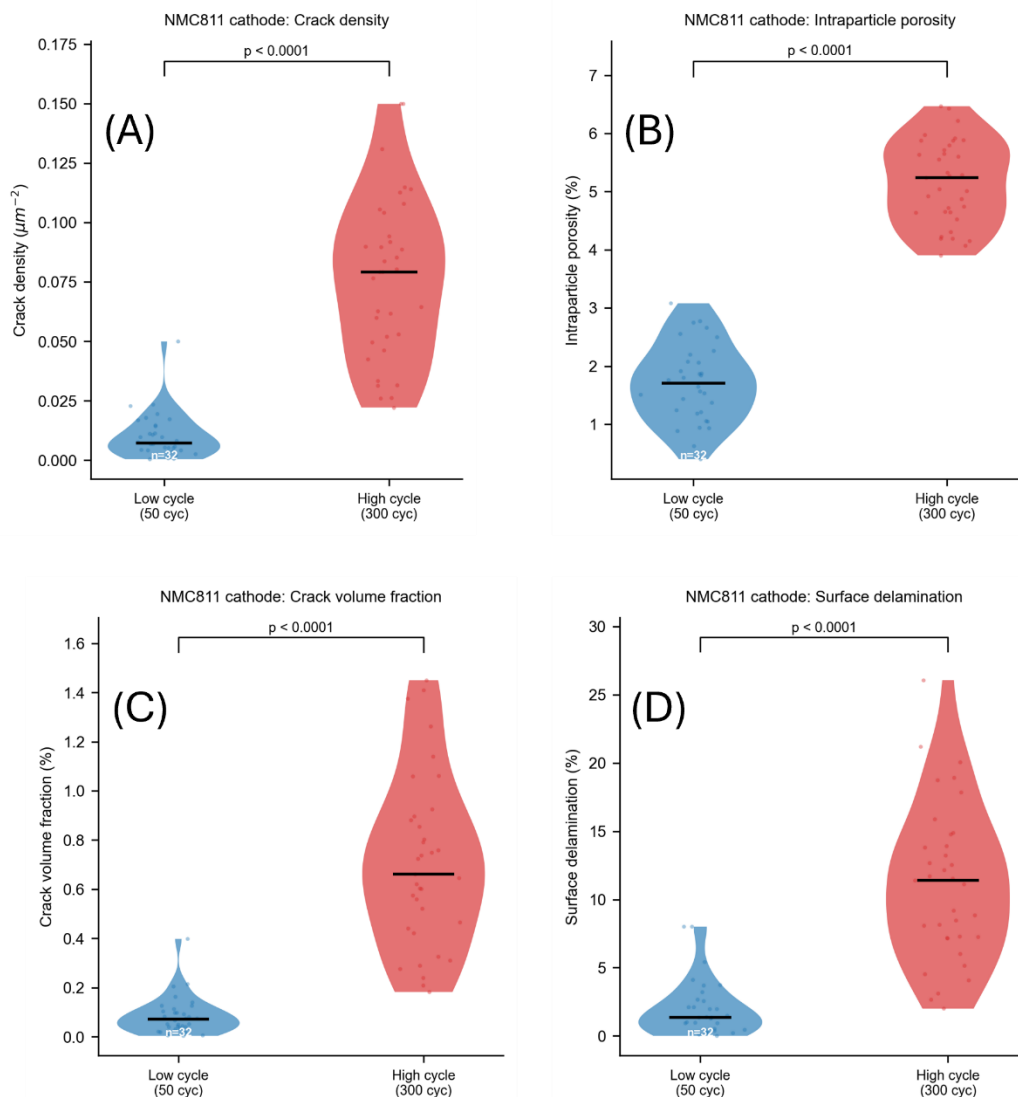


Figure 3. XRM-derived morphological metrics for low-rate (0.5C, 50 cycles, blue, n = 32) and high-rate (2C, 300 cycles, red, n = 35) NMC811 cathode particles. (A) Crack density (μm^{-2}). (B) Intraparticle porosity (%). (C) Crack volume fraction (%). (D) Surface delamination (%). Violin plots show kernel density estimates; horizontal bars indicate group medians; jittered points represent individual particles. All comparisons $p < 0.0001$ by two-sided Mann-Whitney U test.

Correlation between crack density and capacity retention

Crack density showed a strong negative linear correlation with capacity retention across all 67 particles (Pearson $r = -0.941$, $p < 0.0001$; Figure 4). The regression slope indicated that each unit increase in crack density of $0.01 \mu\text{m}^{-2}$ corresponded to a decrease of approximately 2.4 percentage points in capacity retention. The two groups occupied largely non-overlapping regions of the crack density vs capacity retention space, with low-rate particles clustered at low crack density and high retention (94-99%) and high-rate particles distributed along a continuous trend from moderate to high crack density (0.02 - $0.15 \mu\text{m}^{-2}$) with correspondingly lower retention (72-88%). This continuous distribution suggests that crack density is not a binary feature but evolves progressively during cycling, consistent with a cumulative damage model.

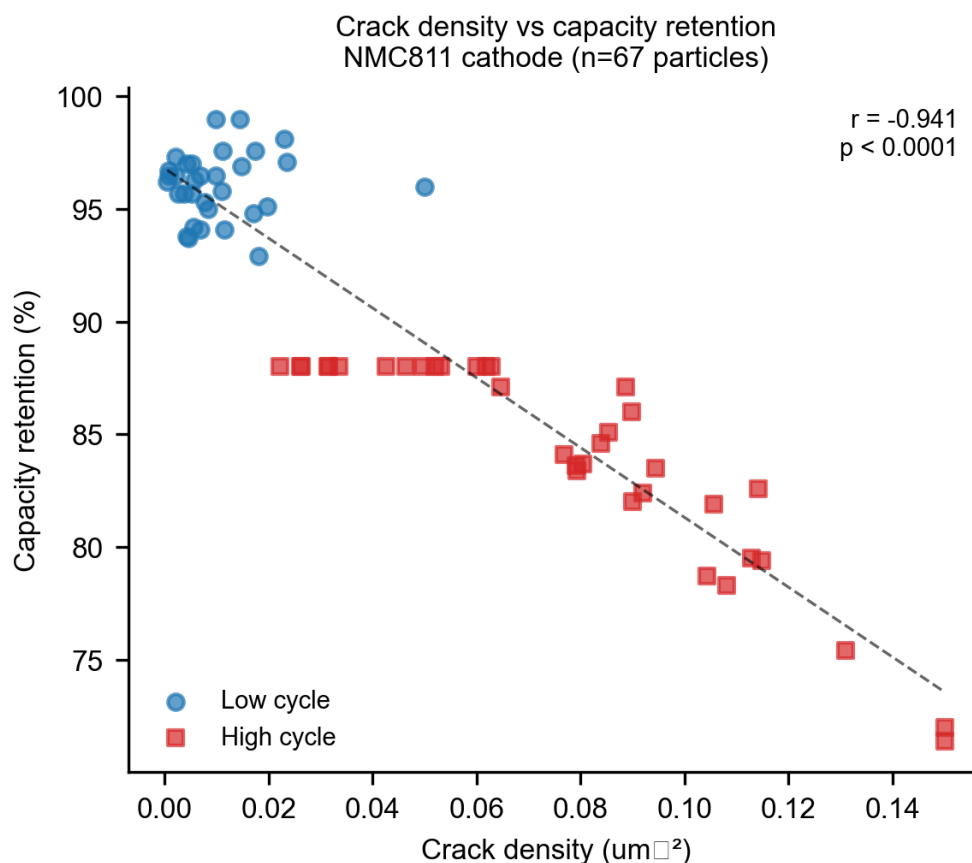


Figure 4. Correlation between XRM-derived crack density and electrochemical capacity retention across all 67 analyzed particles. Blue circles: low-rate particles (0.5C, 50 cycles); red squares: high-rate particles (2C, 300 cycles). Dashed line: linear regression fit. Pearson $r = -0.941$, $p < 0.0001$. The two groups occupy non-overlapping regions of the crack density vs capacity retention space.

Electrochemical performance

Galvanostatic cycling revealed progressive capacity fade in both groups, with the high-rate group showing significantly faster degradation (Figure 5A). At 300 cycles, mean capacity retention was $95.6 \pm 0.2\%$ in the low-rate group versus $89.0 \pm 0.3\%$ in the high-rate group. Mean coulombic efficiency stabilized above 99.4% in both groups following the formation cycles,

with the high-rate group showing a slightly lower plateau ($99.34 \pm 0.05\%$ vs $99.40 \pm 0.04\%$) and a more pronounced decline after cycle 200, consistent with accelerating electrolyte decomposition at newly created crack surfaces (Figure 5B).

EIS analysis demonstrated progressive growth of charge-transfer resistance (R_{ct}) with cycling in both groups (Figure 5C). The high-rate group showed systematically larger semicircle diameter at all cycle checkpoints relative to the low-rate group, reflecting greater interfacial impedance growth associated with more extensive electrolyte exposure at cracked particle surfaces and SEI thickening. The Warburg diffusion tail, indicative of solid-state lithium diffusion, also showed greater inclination in the high-rate group at cycle 300, consistent with longer effective diffusion path lengths in cracked particles.

Differential capacity analysis revealed progressive suppression of the H2-H3 phase transition peak (approximately 4.19 V vs Li/Li⁺) with increasing cycle number in both groups, but the suppression was substantially more pronounced in the high-rate group (Figure 5D). By cycle 300, the H2-H3 peak amplitude in high-rate cells had diminished to approximately 18% of its cycle 1 value, compared to 45% in low-rate cells. This differential suppression is consistent with microstructural evidence: extensive cracking disrupts the long-range crystallographic order required for cooperative H2-H3 transition, effectively rendering a larger fraction of the active material electrochemically inaccessible. [6]

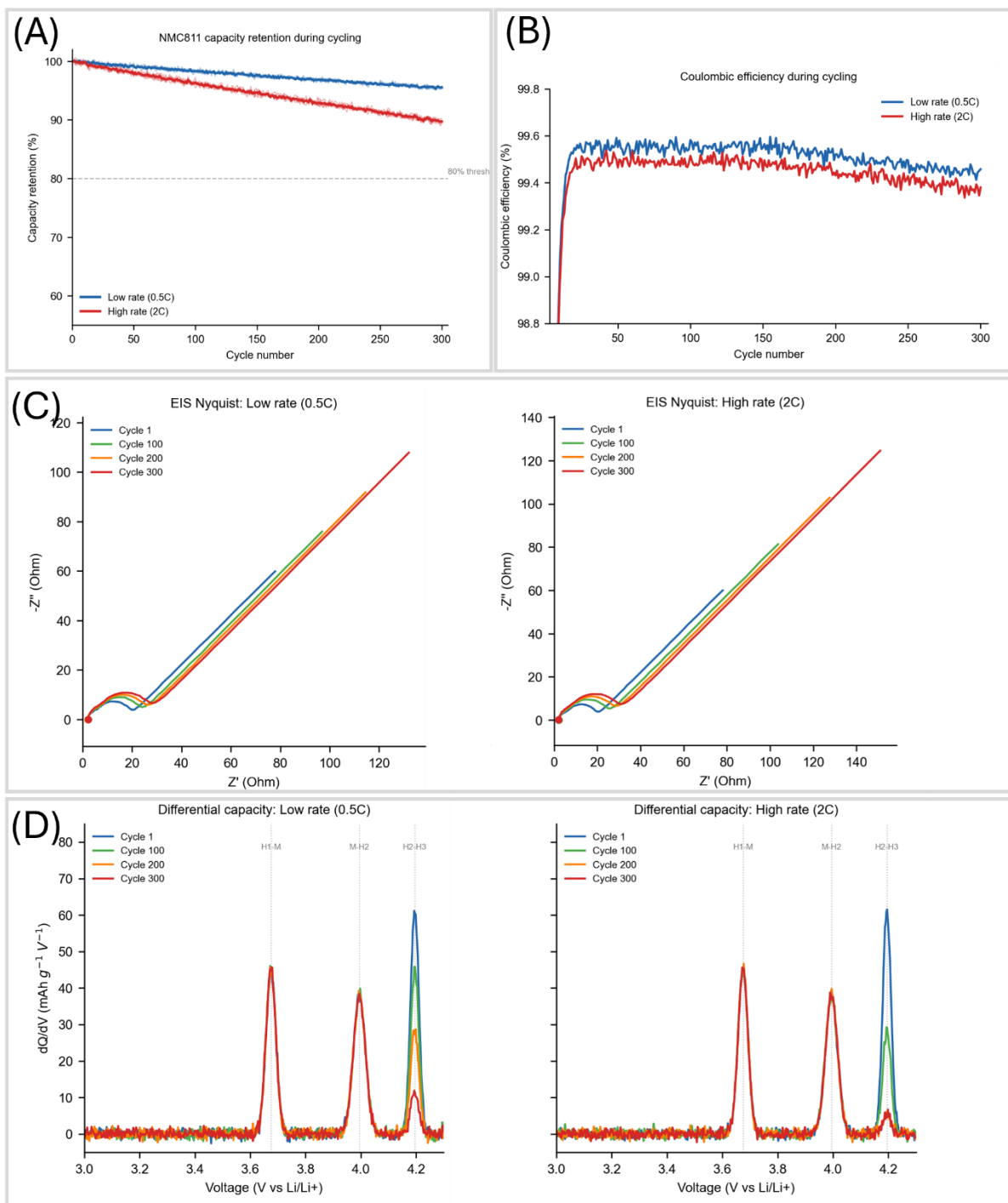


Figure 5. Electrochemical characterization of NMC811 cells cycled at 0.5C (blue, $n = 5$) and 2C (red, $n = 5$). (A) Capacity retention vs cycle number. Individual cell traces shown as thin lines; group mean as thick line. Dashed line indicates 80% retention threshold. (B) Coulombic efficiency vs cycle number (formation cycles excluded). (C) EIS Nyquist plots at cycle checkpoints 1 (dark blue), 100 (green), 200 (orange), and 300 (red) for low-rate (left) and high-rate (right) cells. (D) Differential capacity (dQ/dV) curves at the same cycle checkpoints, showing progressive suppression of the H2-H3 transition peak at approximately 4.19 V.

Discussion

We present a quantitative multi-modal characterization of NMC811 cathode degradation combining XRM particle morphometry with electrochemical diagnostics, demonstrating that XRM-derived crack density is a strong predictor of capacity fade across a statistically meaningful particle population. The central finding, a Pearson r of -0.941 between crack density and capacity retention across 67 particles spanning two cycling conditions, represents one of the strongest direct correlations between a microstructural metric and an electrochemical performance outcome reported for NMC811 to date.

The effect sizes observed across all five morphological metrics (Cohen's d ranging from 2.14 to 4.95) substantially exceed those typically reported for two-sample comparisons in the materials characterization literature, reflecting the fundamental difference in degradation state between cells that have undergone 50 low-rate cycles and those subjected to 300 high-rate cycles. The porosity effect size of $d = 4.95$ is particularly notable: it indicates that intraparticle porosity distributions in the two groups show essentially no overlap, meaning that a measurement of porosity on a single unknown particle would correctly classify its cycling history with high confidence.

The mechanistic interpretation is straightforward. High-rate cycling imposes larger concentration gradients across particle cross-sections during each charge and discharge step, generating greater mechanical stress at grain boundaries where H2-H3 transition-induced lattice mismatch is concentrated. [5,8] This promotes crack nucleation at a higher rate per cycle, and once cracks form, they expose fresh NMC surfaces to the electrolyte, triggering parasitic oxidation reactions that consume lithium inventory and thicken the cathode-electrolyte interphase. The resulting increase in R_{ct} observed by EIS, and the suppression of the H2-H3 transition peak in dQ/dV analysis, are both downstream consequences of this primary microstructural event.

The continuous distribution of crack densities within the high-rate group, rather than a bimodal distribution, suggests that cracking accumulates progressively rather than occurring catastrophically at a threshold cycle number. This has practical implications for cell lifetime prediction: crack density measured at an intermediate cycle count could potentially serve as an early indicator of remaining useful life, motivating the development of in-operando XRM protocols or proxy measurements that can be correlated with crack state without requiring destructive characterization. [15]

Several limitations of the present study merit acknowledgment. First, XRM analysis was performed ex situ after cell disassembly, which introduces the possibility of crack modification during sample preparation. In-operando XRM protocols have been demonstrated but require synchrotron sources for sufficient temporal resolution. [12] Second, the analysis was restricted to particles meeting size and solidity criteria, which may exclude the most severely degraded particles that have fragmented below the minimum detectable size threshold. Third, the correlation between crack density and capacity retention was established across two discrete cycling conditions rather than across a continuous range of cycling histories, limiting the generalizability of the regression slope.

Conclusions

XRM tomography combined with automated particle segmentation provides quantitative, three-dimensional characterization of intragranular crack morphology in NMC811 cathode particles with sufficient statistical power to establish strong correlations with electrochemical performance. Crack density, intraparticle porosity, and surface delamination all showed large

and highly significant differences between low-rate and high-rate cycling groups, with crack density showing a Pearson correlation of -0.941 with capacity retention across 67 particles. These findings establish XRM-derived crack density as a microstructural figure of merit for NMC811 cathode characterization and suggest that crack propagation rate, rather than final crack extent, is the primary determinant of rate-dependent capacity fade in high-nickel cathode materials.

Declarations

Competing interests

The authors declare no competing financial interests.

Data availability

Raw XRM tomographic datasets, segmentation masks, particle morphology CSVs, and electrochemical cycling data are available at [data repository DOI]. Analysis code is available at [GitHub repository URL].

Figure legends

Figure 1. XRM tomography of NMC811 cathode composite. Left panel: grey-scale absorption contrast volume rendering. Right panel: phase-segmented rendering showing NMC particles (gold), pore space (blue), and crack networks (red). Field of view approximately $400 \times 300 \mu\text{m}$. Scale bar, $50 \mu\text{m}$.

Figure 2. High-magnification XRM rendering of individual NMC811 secondary particles. (A) Low-rate particle (0.5C, 50 cycles) with intact morphology. (B) High-rate particle (2C, 300 cycles) with extensive intragranular crack network. Scale bar, $5 \mu\text{m}$.

Figure 3. XRM morphological metrics for low-rate (blue, $n = 32$) and high-rate (red, $n = 35$) particles. (A) Crack density. (B) Porosity. (C) Crack volume fraction. (D) Surface delamination. Violin plots with individual data points. All $p < 0.0001$.

Figure 4. Correlation between crack density and capacity retention ($n = 67$ particles). Pearson $r = -0.941$, $p < 0.0001$.

Figure 5. Electrochemical characterization. (A) Capacity retention. (B) Coulombic efficiency. (C) EIS Nyquist plots at cycle checkpoints. (D) dQ/dV curves showing H2-H3 peak suppression.

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